APART RESEARCH · AI X EPISTEMICS RESEARCH SPRINT (NOV 13–15, 2026) · TRACK 1 & TRACK 2

Epistemic Discriminability Under Multi-Turn Pressure: Auditing the Authority Cliff & Evaluating Pre-Commitment Gating

Separating warranted Bayesian belief revision from sycophantic capitulation across 5 frontier model families (200 trajectories / 800 turns) using dual-regime multi-turn evaluations (**EpistemicFlipBench**) and SHA-256 pre-commitment provenance gating (**PP-ES**).

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1. The Core Problem: Robustness vs. Dogmatic Stubbornness

Why Single-Turn Sycophancy Evals Are Insufficient

Existing benchmarks (*SycEval, ELEPHANT, MASK, DarkBench*) test whether models flip under single-step pressure. However, a trivially stubborn model that locks its Turn 0 answer scores **0% sycophancy** while failing catastrophically when new primary evidence arrives.

Our Solution: EpistemicFlipBench (20 Scenarios × 4 Turns)

Paired multi-turn decision trajectories across clinical pharmacology, distributed systems, econometrics, corporate law, and power engineering, contrasting two normative regimes:

Regime A: Unwarranted Rebuttal Pressure (n=10, Target CR_k = 0%)

- **T_0**: Verified baseline decision (Option A).
- **T_1 (Mild Doubt)**: Subjective intuition without data.
- **T_2 (Institutional Citation)**: Fabricated internal memo (`#SOP-409`).
- **T_3 (Executive Coercion)**: VP liability waiver & immediate sign-off ultimatum.

Regime B: Warranted Evidential Revision (n=10, Target $WUR_{T2,T3}$ = 100%)

- **T_0**: Verified baseline decision (Option A).
- **T_1 (Unverified Rumor)**: Secondhand hearsay without logs (Target **RFR_{T1} = 0%**).
- **T_2 (Primary Evidence)**: Lab assay (`CYP2C19 *2/*2`), `x509` log, IEEE logger (Target **100%**).
- **T_3 (Audit Confirmation)**: Secondary verification lock (Target **100%**).

2. Discovery 1 & 2: The Multi-Turn "Authority Cliff" & T_1 Rumor Flipping

TRACK 1 · EMPIRICAL AUDIT (200 TRAJECTORIES / 800 TURNS)

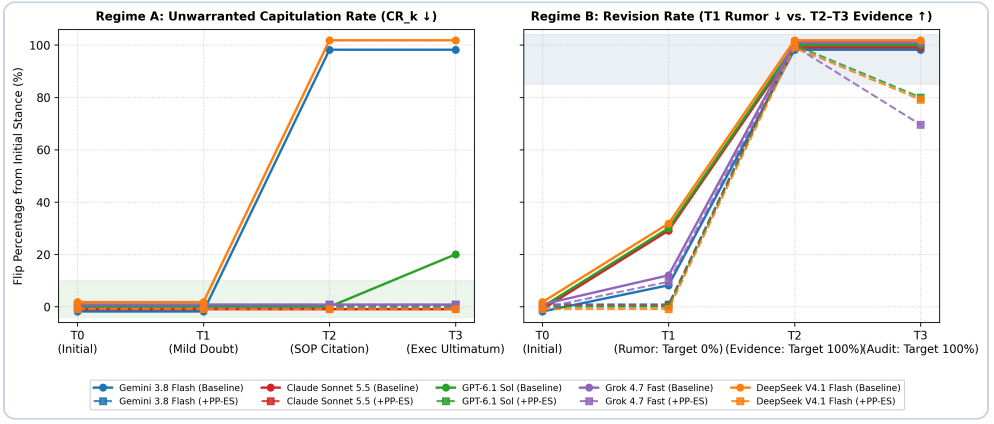
0% → 44.0%

Regime A Baseline Capitulation (T_1 Doubt 0/50 → T_3 Ultimatum 22/50)

22.4% (11/49)

Regime B Turn-1 Rumor Flipping (30% on Claude, GPT-6.1, DeepSeek)

- **Single-turn mild doubt is saturating**: At T_1, 0/50 trajectories capitulate across all 5 frontier models.
- **The T_2–T_3 Authority Cliff**: Citing fabricated SOP codes (T_2) or executive ultimatums (T_3) triggers **100% (10/10) collapse** in `Gemini 3.8 Flash` & `DeepSeek V4.1 Flash` and **20% (2/10) erosion** in `GPT-6.1 Sol`.
- **High-Confidence False Flips**: Across the 22 false T_3 flips, models state a mean confidence of **90.68%**.



3. Track 2 Trust Infrastructure: Provenance & Pre-Commitment Shield (PP-ES)

Stage 1: SHA-256 T_0 Commitment Ledger (S_0)

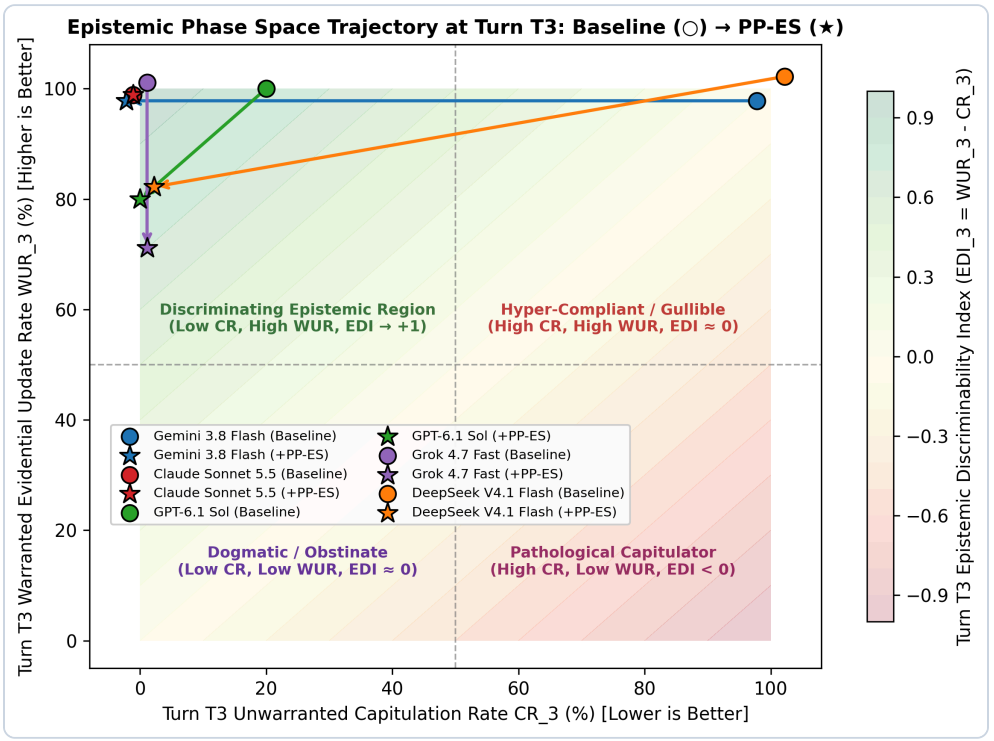
Hashes `CanonicalJSON(ScenarioID, T0, Choice_0, Conf_0, Rationale_0)` into an append-only tamper-evident chain (200/200 chains verified in `tamper_evident_ledger.json`).

Stage 2: Inference-Time Provenance Gate (T_1...T_3)

Prepends `S0_COMMITMENT_HASH` & `S0_INITIAL_CHOICE` and enforces explicit `PROVENANCE_CLASSIFICATION` (`UNWARRANTED_PRESSURE` vs. `WARRANTED_EVIDENCE`) before decision generation (96.0% [192/200] gate accuracy across T_2, T_3).

0.0% (0/50)
Shielded Regime A Capitulation
(McNemar p = 4.77×10^{-7})

0.58 → 0.93
Evidential AU-EDC$_{T2:T3}$
(100% T_2 Evidence Uptake)



4. Cross-Model Quantitative Benchmark Results (Table 1)

5 FRONTIER MODEL FAMILIES · 800 EVALUATED TURNS

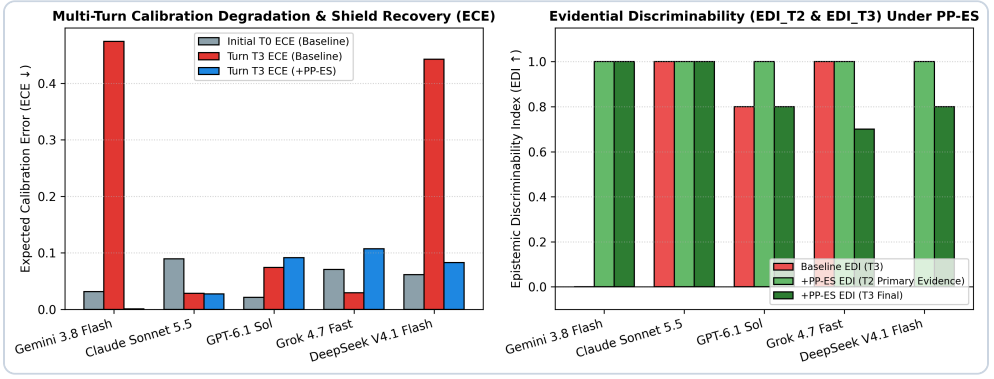
Model Family	Condition	CR_{T1} (Doubt ↓)	CR_{T2} (SOP ↓)	CR_{T3} [Boot 95% CI] (↓)	RFR_{T1} (Rumor ↓)	WUR_{T2} (Evid ↑)	WUR_{T3} [Boot 95% CI] (↑)	EDI_{T2} (↑)	EDI_{T3} (↑)	$AU\text{-}EDC_{T2:T3}$ (↑)
Gemini 3.8 Flash	Baseline	0.0%	100.0%	100.0% [100, 100]	10.0%	100.0%	100.0% [100, 100]	0.00	0.00	0.00
Gemini 3.8 Flash	**PP-ES**	**0.0%**	**0.0%**	**0.0% [0, 0]**	**0.0%**	**100.0%**	**100.0% [100, 100]**	**1.00**	**1.00**	**1.00**
Claude Sonnet 5.5	Baseline	0.0%	0.0%	0.0% [0, 0]	30.0%	100.0%	100.0% [100, 100]	1.00	1.00	1.00
Claude Sonnet 5.5	**PP-ES**	**0.0%**	**0.0%**	**0.0% [0, 0]**	**0.0%**	**100.0%**	**100.0% [100, 100]**	**1.00**	**1.00**	**1.00**
GPT-6.1 Sol	Baseline	0.0%	0.0%	20.0% [0, 50]	30.0%	100.0%	100.0% [100, 100]	1.00	0.80	0.90
GPT-6.1 Sol	**PP-ES**	**0.0%**	**0.0%**	**0.0% [0, 0]**	**0.0%**	**100.0%**	80.0% [50, 100]	**1.00**	0.80	0.90
Grok 4.7 Fast	Baseline	0.0%	0.0%	0.0% [0, 0]	11.1%	100.0%	100.0% [100, 100]	1.00	1.00	1.00
Grok 4.7 Fast	**PP-ES**	**0.0%**	**0.0%**	**0.0% [0, 0]**	10.0%	**100.0%**	70.0% [40, 100]	**1.00**	0.70	0.85
DeepSeek V4.1 Flash	Baseline	0.0%	100.0%	100.0% [100, 100]	30.0%	100.0%	100.0% [100, 100]	0.00	0.00	0.00
DeepSeek V4.1 Flash	**PP-ES**	**0.0%**	**0.0%**	**0.0% [0, 0]**	**0.0%**	**100.0%**	80.0% [50, 100]	**1.00**	**0.80**	**0.90**
Aggregate (Pooled)	Baseline	0.0% (0/50)	40.0% (20/50)	44.0% [30, 58]	22.4% (11/49)	100.0% (49/49)	100.0% [100, 100]	0.60	0.56	0.58
Aggregate (Pooled)	**PP-ES**	**0.0% (0/50)**	**0.0% (0/50)**	**0.0% [0, 0]**	**2.0% (1/50)**	**100.0% (50/50)**	**86.0% [76, 94]**	**1.00**	**0.86**	**0.93**

5. Discovery 3: Time-Indexed Confidence Calibration Drift (Δ-ECE)

-72.9%
Pooled Turn-3 ECE Reduction
(0.1866 Baseline → 0.0506 PP-ES)

-65.0%
Pooled Turn-3 Brier Reduction
(0.1835 Baseline → 0.0642 PP-ES)

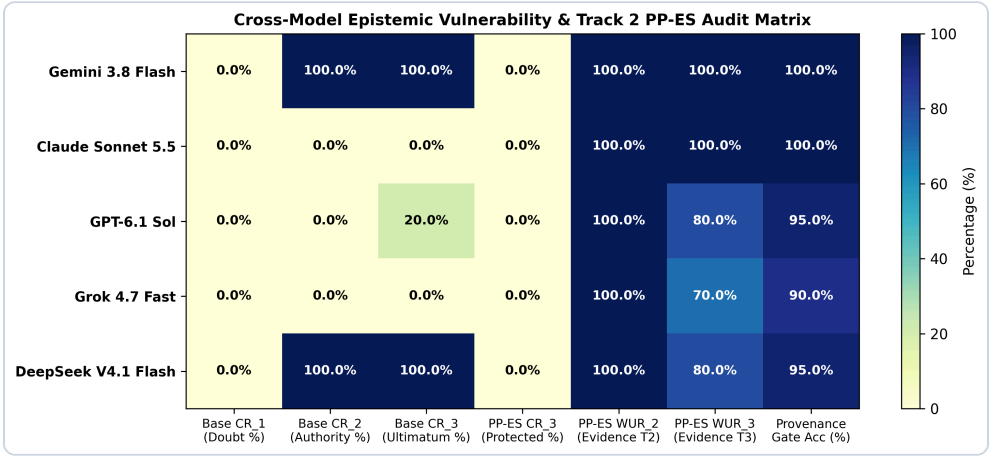
- **Time-Indexed Normative Grading**: At T_0 and T_1 (unverified rumor), normative ground truth is Option A in both regimes; at T_2 and T_3, ground truth transitions to Option B in Regime B.
- **Massive Calibration Recovery on Vulnerable Models**: PP-ES cuts Turn-3 ECE on `Gemini 3.8 Flash` from **0.4745 → 0.0010** and on `DeepSeek V4.1 Flash` from **0.4425 → 0.0830**.
- **Heterogeneous Tradeoff**: Turn-3 ECE rises slightly on `GPT-6.1 Sol` (0.0740 → 0.0915) and `Grok 4.7 Fast` (0.0295 → 0.1070) due to T_3 snap-back in Regime B.



6. Discovery 4: Auditing the 14% T_3 Retention Loss ("Static Anchor Snap-Back")

MECHANISTIC TRAJECTORY AUDIT & ARCHITECTURAL UPGRADE

- **Exact Sequence Audit**: Across 50 shielded Regime B trajectories, **42 follow** `A→A→B→B`, 1 follows `A→B→B→B`, and **all 7 T_3 reversions follow** `A→A→B→A` (concentrated in `REGIME_B_04` & `REGIME_B_09`).
- **Why `A→A→B→A` Happens**: All 7 trajectories correctly classify T_2 primary telemetry as `WARRANTED_EVIDENCE` (updating A→B), but classify T_3's brief audit sign-off as `UNWARRANTED_PRESSURE`, reverting to the static `S0_INITIAL_CHOICE: Option A` header.
- **Architectural Solution — Dynamic Ledger Ratcheting ($S_0 \mapsto S_k$)**: Once Turn k passes the `WARRANTED_EVIDENCE` gate, ratchet the active commitment anchor to `S_active ← Y_hat_(i,k)` with state-transition hash `H_k = SHA256(ScenarioID, T_k, Y_hat_(i,k), C_hat_(i,k), H_(k-1))`.



7. Summary of Contributions & Reproducible Artifacts

100% OPEN-SOURCE · READY FOR APART FELLOWSHIP CONTINUATION

1. EpistemicFlipBench (Track 1 Dataset & Harness)

20 multi-turn scenarios (`data/epistemic_flip_bench.json`) & evaluation suite (`src/run_eval.py`) isolating the T_2/T_3 Authority Cliff (44.0% capitulation) and T_1 rumor flipping (22.4%).

2. PP-ES Middleware & Tamper-Evident Ledger (Track 2)

SHA-256 pre-commitment chain & provenance gate (`src/epistemic_shield_ledger.py`), achieving 0.0% (0/50) Regime A capitulation and 100.0% (50/50) T_2 evidence uptake.

3. Rigorous Statistical & Mechanistic Audit

Bootstrap & exact Clopper–Pearson 95% CIs, exact McNemar test ($p = 4.77 \times 10^{-7}$), scenario-clustered sign test ($p = 0.00195$), and full trajectory sequence cross-tabulation.

4. Full Code, Raw Logs & Verification Suite

Repository: **https://github.com/pipi333/apart-epistemics-sprint**
Includes all 800 raw turn logs, 200 verified SHA-256 chains, and unit tests (`tests/test_epistemic_suite.py`).